



ELECTRICAL MEASUREMENTS

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ELECTRICAL MEASUREMENTS

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads



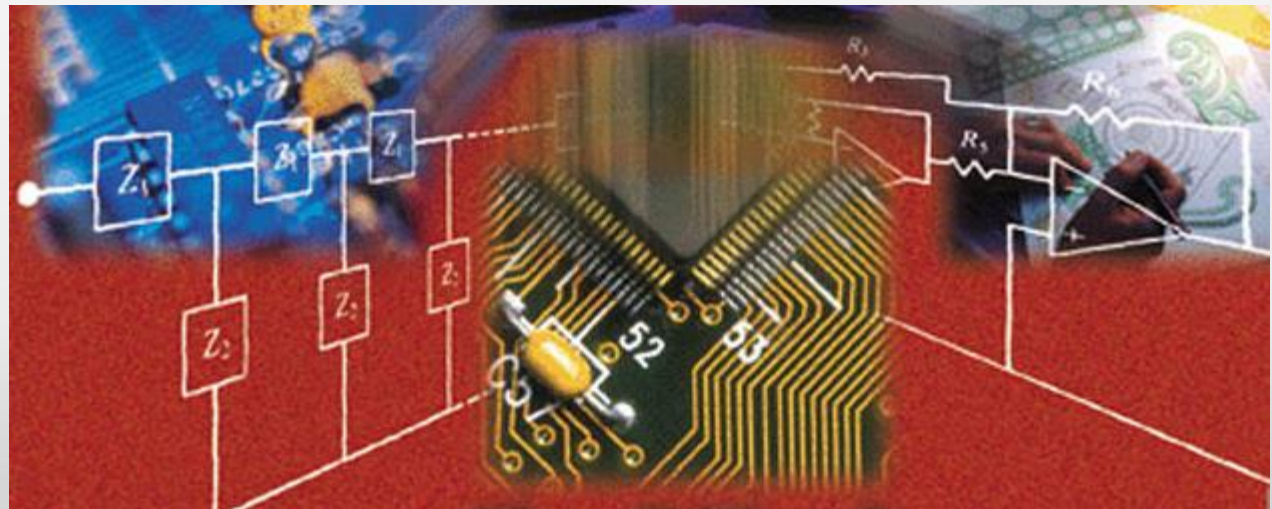
Menoufia University FACULTY OF Electronic Eng

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors



Thanks & Questions

Reduction of systematic errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

- 1. Careful instrument design**
- 2. Method of opposing inputs**
- 3. High-gain feedback**
- 4. Calibration**
- 5. Manual correction of output reading**
- 6. Intelligent instruments**

Reduction of systematic errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

1. Careful instrument design

- ✓ Careful instrument design is the most useful weapon in the battle against environmental inputs, by reducing the sensitivity of an instrument to environmental inputs to as low a level as possible.
- ✓ For instance, in the design of strain gauges, the element should be constructed from a material whose resistance has a very low temperature coefficient (i.e. the variation of the resistance with temperature is very small).

Reduction of systematic errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

2. Method of opposing inputs

- ✓ The method of opposing inputs compensates for the effect of an environmental input in a measurement system by introducing an equal and opposite environmental input that cancels it out.

Reduction of systematic errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

3. High-gain feedback

- ✓ Thus, high gain feedback techniques are often a very effective way of reducing a measurement system's sensitivity to environmental inputs. However, one potential problem that must be mentioned is that there is a possibility that high-gain feedback will cause **instability** in the system.
- ✓ Therefore, any application of this method must include careful stability analysis of the system.

Reduction of systematic errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

4. Calibration

- ✓ All instruments suffer drift in their characteristics, and the rate at which this happens depends on many factors, such as the environmental conditions in which instruments are used and the frequency of their use.
- ✓ Thus, errors due to instruments being out of calibration can usually be rectified by increasing the frequency of recalibration.

Reduction of systematic errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

5. Manual correction of output reading

- ✓ In the case of errors that are due either to system disturbance during the act of measurement or due to environmental changes, a good measurement technician can substantially reduce errors at the output of a measurement system by calculating the effect of such systematic errors and making appropriate correction to the instrument readings.
- ✓ This procedure is carried out automatically by intelligent instruments.

Reduction of systematic errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

6. Intelligent instruments

- ✓ Intelligent instruments contain extra sensors that measure the value of environmental inputs and automatically compensate the value of the output reading.
- ✓ They have the ability to deal very effectively with systematic errors in measurement systems, and errors can be attenuated to very low levels in many cases.

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

2. Random errors are perturbations of the measurement either side of the true value caused by random and unpredictable effects, such that positive errors and negative errors occur in approximately equal numbers for a series of measurements made of the same quantity, when measurements are taken by human observation of an analogue meter and Electrical noise.

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

➤ Statistical analysis of measurements subject to random errors

1. Mean and median values

$$x_{\text{mean}} = \frac{x_1 + x_2 + \cdots x_n}{n}$$

2. Standard deviation and variance

$$d_i = x_i - x_{\text{mean}}$$

$$V = \frac{d_1^2 + d_2^2 \cdots d_n^2}{n - 1}$$

$$\sigma = \sqrt{V} = \sqrt{\frac{d_1^2 + d_2^2 \cdots d_n^2}{n - 1}}$$

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

➤ Statistical analysis of measurements subject to random errors

1. Mean and median values

398 420 394 416 404 408 400 420 396 413 430 (Measurement set A)

mean = 409.0

409 406 402 407 405 404 407 404 407 407 408 (Measurement set B)

mean = 406.0

409 406 402 407 405 404 407 404 407 407 408 406 410 406 405 408

406 409 406 405 409 406 407 (Measurement set C)

mean = 406.5

Thanks & Questions

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

➤ Statistical analysis of measurements subject to random errors

2. Standard deviation and variance

Calculate σ and V for measurement sets A, B and C above.

First, draw a table of measurements and deviations for set A (mean = 409 as calculated earlier):

Measurement	398	420	394	416	404	408	400	420	396	413	430
Deviation from mean	-11	+11	-15	+7	-5	-1	-9	+11	-13	+4	+21
(deviations) ²	121	121	225	49	25	1	81	121	169	16	441

$\sum(\text{deviations}^2) = 1370$; $n = \text{number of measurements} = 11$.

Then, from (3.7), $V = \sum(\text{deviations}^2)/(n - 1) = 1370/10 = 137$; $\sigma = \sqrt{V} = 11.7$.

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

➤ Statistical analysis of measurements subject to random errors

2. Standard deviation and variance

Calculate σ and V for measurement sets A, B and C above.

The measurements and deviations for set B are (mean = 406 as calculated earlier):

Measurement	409	406	402	407	405	404	407	404	407	407	408
Deviation from mean	+3	0	-4	+1	-1	-2	+1	-2	+1	+1	+2
(deviations) ²	9	0	16	1	1	4	1	4	1	1	4

From this data, using (3.7) and (3.8), $V = 4.2$ and $\sigma = 2.05$.

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

➤ Statistical analysis of measurements subject to random errors

2. Standard deviation and variance

Calculate σ and V for measurement sets A, B and C above.

The measurements and deviations for set C are (mean = 406.5 as calculated earlier):

Measurement	409	406	402	407	405	404	407	404
Deviation from mean	+2.5	-0.5	-4.5	+0.5	-1.5	-2.5	+0.5	-2.5
(deviations) ²	6.25	0.25	20.25	0.25	2.25	6.25	0.25	6.25

Measurement	407	407	408	406	410	406	405	408
Deviation from mean	+0.5	+0.5	+1.5	-0.5	+3.5	-0.5	-1.5	+1.5
(deviations) ²	0.25	0.25	2.25	0.25	12.25	0.25	2.25	2.25

Measurement	406	409	406	405	409	406	407
Deviation from mean	-0.5	+2.5	-0.5	-1.5	+2.5	-0.5	+0.5
(deviations) ²	0.25	6.25	0.25	2.25	6.25	0.25	0.25

From this data, using (3.7) and (3.8), $V = 3.53$ and $\sigma = 1.88$.

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

➤ Statistical analysis of measurements subject to random errors

2. Standard deviation and variance

Calculate σ and V for measurement sets A, B and C above.

- Thus, as V and σ decrease for a measurement set, we are able to express greater confidence that the calculated mean or median value is close to the true value, i.e. that the averaging process has reduced the random error value close to zero.
- Comparing V and σ for measurement sets B and C, V and σ get smaller as the number of measurements increases, confirming that confidence in the mean value increases as the number of measurements increases.

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

➤ Statistical analysis of measurements subject to random errors

We have observed so far that random errors can be reduced by taking the average (mean or median) of a number of measurements. However, although the mean value is close to the true value, it would only become exactly equal to the true value if we could average an infinite number of measurements.

Thanks & Questions

Random errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

➤ Statistical analysis of measurements subject to random errors

As we can only make a finite number of measurements in a practical situation, the average value will still have some error. This error can be quantified as the *standard error of the mean*, which will be discussed in detail a little later. However, before that, the subject of graphical analysis of random measurement errors needs to be covered.

Aggregation of measurement system errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

➤ Combined effect of systematic and random errors

If a measurement is affected by both systematic and random errors that are quantified as $\pm x$ (systematic errors) and $\pm y$ (random errors), some means of expressing the combined effect of both types of error is needed. $e = \pm(x + y)$

However, a more usual course of action is to express the likely maximum error as follows:

$$e = \sqrt{(x^2 + y^2)}$$

Aggregation of measurement system errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

➤ Combined effect of systematic and random errors

Error in a sum

$$S_{\max} = (y + ay) + (z + bz); \quad S_{\min} = (y - ay) + (z - bz); \quad \text{or } S = y + z \pm (ay + bz)$$

$$e = \sqrt{(ay)^2 + (bz)^2}$$

$$S = (y + z) \pm e$$

$$S = (y + z)(1 \pm f) \quad \text{where } f = e/(y + z)$$

Thanks & Questions

Aggregation of measurement system errors

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

➤ Combined effect of systematic and random errors

Example 3.6

A circuit requirement for a resistance of $550\ \Omega$ is satisfied by connecting together two resistors of nominal values $220\ \Omega$ and $330\ \Omega$ in series. If each resistor has a tolerance of $\pm 2\%$, the error in the sum calculated according to equations (3.21) and (3.22) is given by:

$$e = \sqrt{(0.02 \times 220)^2 + (0.02 \times 330)^2} = 7.93; \quad f = 7.93/550 = 0.0144$$

Thus the total resistance S can be expressed as

$$S = 550\ \Omega \pm 7.93\ \Omega \text{ or } S = 550 (1 \pm 0.0144)\ \Omega, \quad \text{i.e. } S = 550\ \Omega \pm 1.4\%$$

Thanks & Questions

Questions and Discussion

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions



End of Lecture

Scope of presentation

Introduction

Sources of systematic error

1. System disturbance due to measurement
2. Errors due to environmental inputs
3. Wear in instrument components
4. Connecting leads

Reduction of systematic errors

Random errors

Statistical analysis of measurements subject to random errors

Aggregation of measurement system errors

Thanks & Questions

Good Luck
See You in
Next Lecture . . .



With My Best Wishes
Dr. Eng. Mohamed Abdo